

JATIN MALLARAPU

AI / Machine Learning Engineer

📍 Open to Relocate | 📞 +1(806)-544-0585 | ✉️ jatinmallarapu27@gmail.com | 🔗 linkedin.com/in/jatinmallarapu
🐙 github.com/jatinmallarapu | 🌐 [jatinmallarapu/portfolio](https://jatinmallarapu.com/portfolio)

Professional Summary

AI/ML Engineer with 4+ years of experience designing, building, and deploying production-grade machine learning and generative AI systems across banking, healthcare, telecom, and fintech domains. Proven track record of reducing model deployment time by 35%, improving prediction accuracy by up to 18%, and cutting infrastructure costs by 20% through automated MLOps pipelines. Deep expertise in AWS, Azure, GCP, Docker, Kubernetes, CI/CD, and MLflow, with a strong focus on scalable model monitoring, Generative AI, and enterprise-grade AI infrastructure.

Technical Skills

Languages: Python, SQL, Bash/Shell Scripting, PySpark, Java, Scala

Machine Learning: Scikit-learn, XGBoost, LightGBM, Supervised & Unsupervised Learning, Feature Engineering, Model Optimization, Hyperparameter Tuning

Deep Learning: TensorFlow, PyTorch, Keras, CNN, RNN, LSTM, Transformers

Generative AI & LLMs: LLMs, Hugging Face, LangChain, OpenAI APIs, Prompt Engineering, Fine-Tuning, PEFT/LoRA, Quantization, Distributed Training

Agentic AI: AI Agents, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), RAG Architecture, Vector Search, Semantic Search, Knowledge Graphs

NLP & Computer Vision: BERT, GPT, spaCy, NLTK, Named Entity Recognition (NER), Sentiment Analysis, OpenCV, YOLO, Object Detection, OCR

Vector Databases: FAISS, Pinecone

Data Engineering: Apache Spark, ETL/Data Pipelines, Apache Airflow, Data Warehousing, Big Data Technologies

API & Deployment: FastAPI, Flask, REST APIs, Model Serving, Real-Time & Batch Inference, Docker, Kubernetes, Microservices, Event-Driven Architecture

MLOps & LLMOps: MLflow, Kubeflow, CI/CD Pipelines, Feature Store, Model Registry, Data Versioning, Model Monitoring, A/B Testing

Cloud Platforms: AWS (SageMaker, S3, Lambda), Google Cloud Platform (Vertex AI), Microsoft Azure (Azure ML)

Databases: MySQL, PostgreSQL, MongoDB, NoSQL Databases

Visualization & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI, Jupyter Notebook

Governance & Leadership: Explainable AI (XAI), Responsible AI, AI Governance, System Design, Solution Architecture, Agile/Scrum, Stakeholder Management, Technical Leadership

Professional Experience

Senior AI/ML Engineer

Sep 2025 – Present

Client: U.S. Bank, Irving, TX

- Developed scalable AI/ML solutions using Python, TensorFlow, PyTorch, Scikit-learn, XGBoost, and Apache Spark for banking risk analytics and fraud detection, improving fraud-flag precision by 22%.
- Built and optimized Supervised, Unsupervised, and Deep Learning models using Feature Engineering and Hyperparameter Tuning, lifting model accuracy by 15% across risk-scoring use cases.
- Designed Generative AI applications using LLMs, Transformers, Hugging Face, LangChain, and PEFT/LoRA for intelligent document processing, cutting manual review time by 40%.
- Developed RAG Architecture, Agentic AI, and Multi-Agent Systems with Vector Databases (FAISS, Pinecone), reducing enterprise knowledge-retrieval latency by 30%.
- Deployed scalable Flask and REST API services with Docker and Kubernetes supporting Real-Time and Batch Inference for over 1M+ daily transactions.
- Implemented MLOps/LLMOps pipelines using MLflow, Kubeflow, and CI/CD Pipelines, reducing model deployment cycle time from weeks to under 48 hours.

- Processed large-scale banking datasets using Apache Spark, ETL Pipelines, and Apache Airflow across AWS (SageMaker, S3, Lambda) and GCP (Vertex AI), improving pipeline throughput by 25%.
- Applied Explainable AI (XAI) and Responsible AI practices to strengthen model governance and regulatory compliance across all production deployments.
- Collaborated in Agile/Scrum environments on System Design and Solution Architecture, driving end-to-end delivery of enterprise-scale AI solutions.

Machine Learning Engineer

Feb 2025 – Aug 2025

Client: Steward Health Care, Dallas, TX

- Designed Machine Learning and Deep Learning solutions using Python, TensorFlow, PyTorch, and XGBoost to improve medication adherence prediction, boosting model accuracy by 17%.
- Engineered scalable ETL and Data Pipelines using Apache Spark, PySpark, and Pandas for high-volume clinical datasets, reducing pipeline runtime by 28%.
- Built Generative AI applications using LLMs, Hugging Face, LangChain, and RAG Architecture to automate clinical document search, saving an estimated 10+ hours/week of manual lookup.
- Developed secure FastAPI, Flask, and REST API services with Docker and Kubernetes for Real-Time and Batch Inference in enterprise AI applications.
- Automated model lifecycle management using MLflow, Kubeflow, Apache Airflow, and CI/CD Pipelines, cutting model release time by 30%.
- Leveraged Microsoft Azure (Azure ML) to train and deploy models while applying Explainable AI (XAI) and A/B Testing to validate a 12% lift in production performance.
- Partnered with healthcare stakeholders in Agile/Scrum environments on System Design and Solution Architecture for pharmacy-focused AI solutions.

Data Scientist

Jun 2023 – Jun 2024

Client: BT Group, Bangalore, India

- Analyzed large-scale telecom datasets using Python, SQL, Pandas, and Scikit-learn, improving customer churn prediction accuracy by 19%.
- Built predictive models using XGBoost, LightGBM, Logistic Regression, and Random Forest with Feature Engineering, driving a 14% increase in customer segmentation precision.
- Developed Time Series Forecasting solutions with Apache Spark, ETL Pipelines, and Apache Airflow to forecast network demand, improving capacity planning accuracy by 20%.
- Created interactive dashboards using Power BI, Tableau, and Matplotlib, enabling real-time KPI monitoring for 5+ executive stakeholder teams.
- Collaborated with cross-functional teams in Agile/Scrum environments, applying AWS and Data Warehousing best practices to enterprise reporting.

Data Scientist

Jan 2022 – May 2023

Client: Razorpay, Bangalore, India

- Collected and analyzed digital payment data using Python, SQL, Pandas, and MySQL, identifying transaction trends across 2M+ monthly records.
- Built machine learning models using Scikit-learn, XGBoost, and LightGBM for payment success prediction and fraud pattern identification, improving fraud detection rate by 16%.
- Developed PySpark and Apache Spark ETL Pipelines to process high-volume payment records, improving data quality and reporting efficiency by 25%.
- Applied Time Series Forecasting and anomaly detection to monitor payment gateway performance, reducing false-positive fraud alerts by 12%.
- Delivered interactive Power BI and Tableau dashboards while collaborating with product managers in Agile/Scrum teams.

Education

Master of Science in Computer Science

Aug 2024 - May 2026

Texas Tech University, Lubbock, Texas

GPA: 3.8/4.0

Certifications

- AWS Certified Machine Learning – Specialty
- Microsoft Certified: Azure AI Engineer Associate